

Voting Platform

Developer Specification

Product requirements, voter UX, payments, raffle, leaderboards, admin controls and acceptance criteria

Document status	Implementation-ready requirements v1.0
Primary CTA	Vote for Your Favorite ACE Creator
Core voter flow	Find creator -> Sign in / verify -> Choose votes -> Pay -> Vote confirmed
Open configuration	Voting start/end dates; Friday raffle cutoff/announcement time

Product principle

The voter should experience only three simple actions: Find, Verify, Vote. Everything else - account creation, payment confirmation, vote allocation, raffle entry and leaderboard updates - should happen automatically.

1. Executive summary

ACE Awards will use a mobile-first voting platform that minimizes registration friction and keeps the nominee, payment and confirmation context intact from start to finish. A voter can arrive from the homepage, a category page, search, or a creator-specific link.

North-star experience

Find creator -> Quick sign-in / verification -> Choose vote quantity -> Pay with SantimPay -> Vote counted -> Weekly raffle entry created -> Share, vote again, or view leaderboard.

Confirmed product decisions

Area	Decision
Login	Google + Telegram + phone
Phone uniqueness	One voter account per phone number
Session	Persistent login
Phone verification	Not required for every Google/Telegram voter; required for phone route and available as risk step-up
Voting limit	No purchase/vote limit
Base price	1 vote = 10 ETB
Custom amount	Allowed; best applicable package rate; minimum 10 ETB
Payment	SantimPay; one payment applies to one nominee
Vote activation	Only after server-side successful payment confirmation
Raffle	1 entry per successful transaction, regardless of number of votes
Weekly draw	Friday; 8 winners: 7 phone prizes + 1 larger-phone grand prize
Next raffle	Opens Monday
Leaderboard	Rank + approximate public totals; exact totals internal
Ranking visibility	Visible until voting closes
Languages	English + Amharic

2. Product goals and guardrails

Goals

- Make a first vote possible with minimal friction on a mobile device.
- Allow voters to discover nominees by search, category, or direct creator link.
- Keep the selected nominee and intended vote quantity through sign-in and payment.
- Confirm payment on the server before allocating votes.
- Create one weekly raffle entry for every successful voting transaction.
- Make competition momentum visible through category and overall leaderboards.
- Give administrators control without sacrificing auditability.

Non-goals for launch

- Open public nomination during the voting period. Nominees are fixed before voting opens.
- Splitting a single payment across multiple nominees.
- Manual, unaudited changes to vote totals.
- Dependence on a social network API for core nominee or voting availability.

3. End-to-end voter journey

Step	User sees	System does
1. Discover	Search, categories, leading creators, or direct nominee page	Loads curated nominee record; optionally refreshes social profile enrichment
2. Select	Nominee card or profile with Vote button	Stores nominee context in session/deep-link state
3. Sign in	Telegram, Google, or phone	Creates/links voter profile; enforces one account per phone when phone is available
4. Verify if needed	OTP for phone route or risk step-up	Marks verification status; returns voter to selected nominee
5. Choose votes	Preset quantities and custom amount	Calculates vote quantity and payable amount; minimum 10 ETB
6. Pay	Pay with SantimPay	Creates unique order/payment reference and redirects/opens payment flow
7. Confirm	Pending -> success or failed	Backend verifies gateway result; idempotently finalizes transaction
8. Count	Success page	Creates vote ledger entry and updates nominee/category/overall totals

Step	User sees	System does
9. Enter draw	Weekly raffle confirmation	Creates exactly 1 raffle entry for the successful transaction
10. Re-engage	Vote Again, Share, View Leaderboard	Deep-links back to nominee, share targets, or rankings

Critical UX rule

Never send a voter back to search after authentication or payment. Preserve nominee, category, intended vote quantity and return path across every redirect.

4. Screen-by-screen UX specification

4.1 Homepage

- Primary heading: "Vote for Your Favorite ACE Creator".
- Primary search field: search by creator name, @handle, or category.
- Category tiles/cards.
- Leading creators preview from each category.
- Overall Top ACE Creator preview.
- Weekly raffle panel showing current draw window, prizes and previous winners.
- Language toggle: English / Amharic.
- Persistent navigation to Leaderboard, Weekly Draw, My Votes/Profile.

4.2 Search and category results

Each nominee card should display the minimum information needed to recognize and act:

- Profile photo
- Display name
- @handle
- Category or categories
- Current rank
- Approximate vote total (for example, 12.4K+ votes)
- Primary Vote button

Filtering should support category and text search. Search should tolerate partial names and handle prefixes such as @.

4.3 Nominee detail page

- Large profile photo and display name.
- @handle and category badges.
- Current category rank and approximate vote total.
- Optional social-enrichment metrics when data is legally and technically available.
- Primary Vote button.

- Share button and creator-specific URL.
- If nominee appears in multiple categories, clearly show all eligible categories and the leaderboard context.

4.4 Sign-in / voter profile

Option	Expected behavior
Continue with Telegram	Create/link voter identity. If phone permission is available, link phone without forcing a second registration form.
Continue with Google	Create/link voter identity using available profile fields. Do not require a password.
Continue with phone	Collect name + phone, send OTP, verify, create/link voter profile.

Minimum profile fields: display name, phone when available/required, email when available, authentication provider(s), verification status, account creation date, last login and risk flags.

	Account rule One phone number maps to one voter account. If a Google or Telegram account later supplies a phone already attached to another account, the system must enter an account-linking/recovery path rather than silently create a duplicate.
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4.5 Vote selection

Choice	Price	Votes
1 vote	10 ETB	1
2 votes	20 ETB	2
3 votes	30 ETB	3
5 votes	50 ETB	5
10 votes	100 ETB	10
25 votes	250 ETB	25
50 votes	500 ETB	50
100 votes	1,000 ETB	100

Custom amount is allowed with a 10 ETB minimum. At launch, use a simple 10 ETB-per-vote conversion. Only accept amounts that resolve to whole votes. If discounted packages are introduced later, the pricing service must return the best applicable package rate rather than hard-coding arithmetic in the UI.

4.6 Payment

- Show nominee, vote quantity and total before payment.
- One primary button: Pay with SantimPay.
- One payment applies to one nominee only.
- Create a unique immutable internal order ID before redirecting to the gateway.
- Preserve the intended return URL and nominee context.

4.7 Payment status screens

State	UI	Backend rule
Pending	"Confirming your payment..."; safe refresh/poll	Do not allocate votes yet.
Success	Votes counted, nominee rank, raffle entry, CTAs	Show only after server-side confirmed payment and finalized vote ledger.
Failed	Reason where safe + Try Again	Keep nominee and vote selection; create a new payment attempt/reference.
Unknown/timeout	"We are checking your payment"	Reconcile asynchronously/server-side; never assume success from browser redirect.

4.8 Success screen

- Confirmation of number of votes counted and creator supported.
- Updated/most recent known category rank.
- Confirmation: "You have 1 entry in this week's ACE Draw" for this transaction.
- Current total raffle entries for that voter during the active draw.
- Three CTAs: Vote Again, Share, View Leaderboard.
- Offer receipt/confirmation through available channels: on-screen, email, Telegram and/or SMS depending on user data and integration availability.

5. Authentication, identity and verification rules

Rule	Requirement
Supported methods	Telegram, Google, phone + OTP
Password	No ACE password for Google/Telegram sign-in
Phone uniqueness	A verified phone may belong to only one voter account

Rule	Requirement
Persistent session	Keep user signed in until logout/session expiry/security revocation
Phone requirement	Do not block every Google/Telegram vote on phone verification
Step-up verification	System may require phone OTP for suspicious activity, account recovery, duplicate identity resolution or policy changes
Return context	After auth/OTP, return directly to the nominee and vote selection context the user started from

6. Nominee catalog and social-data enrichment

The ACE database is the source of truth for nominee identity, eligibility, category membership and voting status. Social APIs may enrich presentation but must not control whether voting works.

Core nominee fields

- Nominee ID
- Display name
- Primary @handle
- Profile image
- Platform/profile URL
- Category IDs (one or more)
- Short bio (optional)
- Voting status
- Share slug
- Admin sort/display metadata

Social API enrichment

- Where approved APIs and permissions allow, enrich profile photo, handle/display name, follower count and selected public metadata.
- Cache external data and refresh on a controlled schedule; do not call a social API on every voter page request.
- Keep an admin override for every displayed nominee field.
- If an API is unavailable, rate-limited or changes behavior, fall back to the stored ACE nominee record without blocking votes.

7. Pricing, SantimPay and transaction lifecycle

Pricing service

Pricing should be returned by a backend configuration/service so ACE can change quantities or offers without an app release. The UI must never be the authoritative calculator.

Required payment lifecycle

1. Create internal vote order in CREATED status with voter, nominee, requested votes, amount and pricing version.
2. Create SantimPay payment attempt with a unique gateway reference mapped to the internal order.
3. Move order to PAYMENT_PENDING once handed to the gateway.
4. Accept browser return as display/navigation information only.
5. Confirm payment server-to-server using the gateway callback/webhook and/or verified transaction-status lookup supported by the integration.
6. If confirmed paid, atomically mark payment SUCCESS and finalize vote allocation exactly once.
7. Create one raffle entry for the transaction if the active raffle window accepts entries.
8. Emit/update leaderboard projections.
9. If payment fails, mark attempt FAILED and allow Try Again with a new payment attempt while retaining the original vote order context.
10. If payment is later refunded/reversed, create reversing vote-ledger entries and remove/void associated raffle eligibility according to the final raffle rules and draw status.

Idempotency requirement

A duplicate callback, user refresh or repeated status check must never create duplicate votes or duplicate raffle entries. Finalization must be protected by a unique transaction/payment constraint and an idempotent server operation.

8. Vote counting, refunds and auditability

Use an append-only vote ledger rather than directly editing counters as the source of truth. Cached counters may be maintained for speed, but they must be rebuildable from the ledger.

Event type	Effect
PURCHASE_CREDIT	Adds N paid votes to one nominee after confirmed payment.
REFUND_DEBIT	Removes N votes corresponding to a refunded/reversed payment.
ADMIN_ADJUSTMENT	Adds/removes votes only with required reason, actor, timestamp and linked support reference.
RECONCILIATION	Technical correction produced by controlled reconciliation; fully audited.

Admins may flag or cancel suspicious votes. Any manual change requires a reason and permanent audit log. Deleting historical ledger records should not be allowed through normal admin UI.

9. Leaderboard specification

Public leaderboards

- Category leaderboard for every category.
- Overall Top ACE Creator leaderboard across all categories.

- All nominees visible, not only Top 10.
- Ranking visible through the close of voting; no final-hour freeze.
- Update immediately after a finalized successful vote transaction, subject to normal cache propagation.
- Display rank plus approximate totals, for example 12.4K+ votes rather than exact 12,437.
- Admin dashboard always shows exact counts.

Ranking rules

- Primary sort: eligible paid vote total descending.
- Tie-break rule should be deterministic and configurable before launch; recommended default is earliest time reaching the tied total, then stable nominee ID.
- Refunds/reversals immediately recalculate ranking.
- If a nominee appears in multiple categories, votes should be attributed according to the product's final category voting model. If one vote supports the nominee globally, category leaderboards use that same nominee total; if votes are category-specific later, the vote order must store category ID.

Implementation choice to keep flexible

The current requirements allow a nominee to appear in multiple categories. Store category IDs on nominees and make category attribution explicit in the vote/order schema so ACE can support either global nominee votes or category-specific votes without redesigning the database.

10. Weekly ACE Draw / raffle

Weekly cycle

Period	Behavior
Monday	New weekly draw opens; entry counter resets for the new draw window.
Monday-Friday	Each successful eligible voting transaction creates exactly 1 raffle entry.
Friday cutoff	Stop accepting entries for that draw at an admin-configured cutoff time.
Friday draw	System randomly selects 8 distinct winners from eligible entries.
Friday publication	Publish masked winner information after draw completion/approval policy.
Saturday-Sunday	Display winners and upcoming draw; no new weekly draw until Monday.

Prize structure

- 7 standard phone prizes
- 1 grand prize: larger/premium phone

- Prize models/details configurable by admin each week

Entry rule

One successful transaction = one raffle entry, regardless of whether the transaction purchased 1 vote or 100 votes. Multiple separate successful transactions create multiple entries.

Winner selection requirements

- System-generated random selection.
- 8 distinct winners per draw unless the business explicitly changes the rule.
- An account/phone should not win more than once in the same draw by default.
- Store draw seed/method metadata or equivalent reproducibility/audit data where technically appropriate.
- Persist selected winners before publication.
- Public display uses name + masked phone only (for example, Hana M. - 09*****123).
- Voter profile shows current draw entry count and historical wins/entries as appropriate.

Compliance dependency

Keep raffle eligibility, winner selection and prize configuration modular. Before production launch, ACE should validate the paid-entry promotion structure, eligibility rules, disclosures and winner process with appropriate local legal/compliance review.

11. Sharing and deep links

- Every nominee receives a stable, human-readable share URL (for example, /vote/{creator-slug}).
- Links should open directly on the nominee page with Vote as the primary action.
- Share targets: TikTok where supported, Telegram, WhatsApp, Facebook and Copy Link.
- After voting, offer all three actions: Vote Again, Share, View Leaderboard.
- Preserve campaign/referral parameters for analytics without affecting vote validity.

12. Admin dashboard

Recommended modules

Module	Key capabilities
Dashboard	Revenue, successful transactions, votes, active voters, payment success rate, raffle entries, anomalies.
Nominees	View/edit curated nominee records, categories, images, handles, social metadata overrides, share slugs.
Categories	Create/edit ordering, display text and active state before voting opens.

Module	Key capabilities
Voting	Exact totals, transaction drill-down, ledger events, flags, adjustments with mandatory reason.
Payments	Order/payment status, gateway reference, retries, reconciliation, refunds/reversals.
Voters	Profile, authentication methods, verified phone, transaction history, risk flags, account merge/recovery tools.
Leaderboard	Exact category and overall standings; refresh/rebuild controls.
Raffle	Configure draw window, prizes, cutoff, execute draw, view audit data, publish masked winners.
Reports	Revenue, votes sold, votes by nominee/category, payment success/failure, voter count, raffle entries, daily sales.
Audit log	Admin actions, vote adjustments, refunds, configuration changes, raffle execution/publication.

Admin access

A single admin role is acceptable for launch, but every privileged action must record actor identity, timestamp, object changed, before/after values when practical and a required reason for financial/vote-affecting changes. The schema should not prevent multiple roles from being introduced later.

13. Fraud, security and reliability controls

- Unique payment/gateway references and idempotent payment finalization.
- One verified phone per voter account.
- Rate limits on OTP requests, login attempts, search abuse and payment initiation.
- Duplicate payment-reference detection.
- Risk signals for unusual transaction velocity, repeated payment failures, account/phone/device anomalies and abrupt vote spikes.
- Admin flag/review capability for suspicious transactions and voters.
- No client-side authority over paid status, vote totals, raffle entries or winner selection.
- Encrypt sensitive data in transit; restrict and log administrative access.
- Store only data necessary for product, payment, security and compliance needs.
- Automated reconciliation job comparing internal successful payments/votes to gateway records.
- Backups and a tested restore procedure for transactional data.

14. Recommended data model

Entity	Core fields
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Entity	Core fields
Voter	id, display_name, phone, phone_verified_at, email, status, created_at, last_login_at, risk_status
AuthIdentity	id, voter_id, provider, provider_subject_id, provider_metadata, linked_at
Category	id, name_en, name_am, slug, status, display_order
Nominee	id, display_name, handle, image_url, share_slug, status, social_profile_url, metadata
NomineeCategory	nominee_id, category_id
VoteOrder	id, voter_id, nominee_id, category_id nullable, vote_qty, amount_etb, pricing_version, status, created_at
PaymentAttempt	id, order_id, gateway, gateway_reference, amount, status, raw_status_code, created_at, confirmed_at
VoteLedger	id, order_id, nominee_id, category_id nullable, event_type, vote_delta, reason, actor_id nullable, created_at
RaffleDraw	id, week_key, opens_at, closes_at, draw_at, status, prize_config, audit_metadata
RaffleEntry	id, draw_id, voter_id, order_id UNIQUE, created_at, eligibility_status
RaffleWinner	id, draw_id, entry_id, voter_id, prize_tier, selected_at, published_at, status
AdminAudit	id, admin_id, action, object_type, object_id, before_json, after_json, reason, created_at

Database constraints worth enforcing

UNIQUE RaffleEntry(order_id); unique successful gateway reference; unique provider + provider subject; unique verified phone mapping where applicable; transactional finalization of payment + vote ledger + raffle entry.

15. Backend service boundaries and events

Suggested service/API areas

- Auth/identity service: sign-in, OTP, account linking, session management.
- Catalog service: categories, nominees, search, social enrichment cache.
- Pricing service: packages, custom amount validation, pricing version.
- Voting service: order creation, vote ledger, counters, refunds, adjustments.
- Payment integration: SantimPay initialization, callback/status verification, reconciliation.

- Leaderboard service: category and overall aggregates, public approximation formatting.
- Raffle service: draw windows, entry creation, random selection, prize/winner publication.
- Notification service: receipt/confirmation via available SMS/email/Telegram channels.
- Admin/reporting service: dashboard metrics, audit, exports.

Useful domain events

Event	Consumers
payment.confirmed	Voting finalizer, receipt/notifications
vote.finalized	Leaderboard updater, analytics
raffle.entry.created	Voter profile counter, analytics
payment.refunded	Vote reversal, raffle eligibility review, leaderboard
raffle.draw.completed	Winner publication workflow, notifications
admin.vote.adjusted	Leaderboard, audit, analytics

16. Bilingual UX, accessibility and responsive behavior

- All user-facing copy must support English and Amharic from translation resources, not hard-coded strings.
- Language selection should persist across sessions where possible.
- Nominee names/handles should not be machine-translated unless ACE provides approved alternate names.
- Mobile-first layout; large tap targets; payment CTA visible without excessive scrolling.
- Keyboard-accessible controls and meaningful focus states for desktop/web users.
- Images require useful alt text; profile image alt can use nominee display name.
- Color must not be the only indicator of rank, success, failure or selection.
- Provide clear, plain-language payment states and recoverable error messages.

17. Analytics and reporting

Admin reports

- Total revenue
- Successful/failed/pending payments
- Votes sold
- Votes by nominee
- Votes by category
- Daily/weekly transaction volume
- Unique voters
- New vs returning voters
- Payment conversion/success rate
- Raffle entries by week
- Raffle winners and prize status

- Refunds/reversals
- Suspicious/flagged activity

Funnel events

Funnel stage	Suggested event
Discover	home_view, search_used, category_opened, nominee_opened
Intent	vote_clicked, package_selected, custom_amount_entered
Identity	login_method_selected, login_completed, otp_sent, otp_verified
Payment	payment_started, payment_pending, payment_failed, payment_confirmed
Outcome	vote_finalized, raffle_entry_created, share_clicked, vote_again_clicked, leaderboard_viewed

18. Launch acceptance criteria

- A voter arriving on a nominee deep link can complete sign-in, payment and return to that same nominee without re-searching.
- Google, Telegram and phone login routes all create/link a usable voter profile.
- A phone number cannot be verified onto two separate voter accounts without an explicit recovery/linking resolution.
- Vote quantity and payable amount are calculated by backend pricing rules; minimum custom payment is 10 ETB.
- One payment can target only one nominee.
- No votes are allocated from a browser success redirect alone.
- A confirmed payment allocates votes exactly once even if the callback/status endpoint is delivered multiple times.
- A successful transaction creates exactly one raffle entry in the active weekly draw.
- A failed payment retains nominee and quantity and supports Try Again with a new payment attempt.
- A refund/reversal produces a traceable vote reversal and recalculates leaderboard totals.
- Public leaderboard shows every nominee, rank and approximate totals and remains visible until voting closes.
- Admin leaderboard shows exact totals.
- Weekly draw accepts entries Monday-Friday, selects 8 distinct winners and records prize tier.
- Public winners display masked phone information only.
- Admin vote adjustments require a reason and appear in a permanent audit trail.
- English and Amharic UI can be switched without losing the active voter flow.
- Every nominee has a shareable direct voting URL.
- Core voting remains functional if a social enrichment API is unavailable.

19. Configuration values and remaining TBDs

Item	Status / recommendation
Voting opens	TBD - store as admin configuration and display exact date/time to voters.
Voting closes	TBD - store as admin configuration; leaderboard remains visible until close.
Friday raffle cutoff	TBD exact time - configurable per draw.
Friday winner announcement	TBD exact time - configurable per draw.
Raffle timezone	Use the competition/business timezone consistently across DB, UI and draw execution.
Grand prize model	Larger/premium phone; exact model configurable each week.
Standard prize models	7 phones; exact model(s) configurable each week.
Tie-break rule	Recommended: earliest to reach tied total, then stable nominee ID. Confirm before launch.
Vote/category attribution	Keep schema flexible because nominees may appear in multiple categories; confirm whether votes are global per nominee or category-specific.
Notification channels	Multiple; enable based on available contact data and integrations.
Legal/compliance copy	Finalize terms, raffle eligibility/disclosures, privacy notice and refund policy before production.

20. Recommended implementation sequence

Phase	Scope
Phase 1 - Core voting	Nominee catalog/search/categories, auth, pricing, SantimPay integration, verified payment finalization, vote ledger, success/failure flows.
Phase 2 - Rankings	Category + overall leaderboards, approximate public totals, exact admin totals, refund recalculation.
Phase 3 - Weekly draw	Raffle windows, 1 entry/transaction, 8-winner selection, masked winners, prize configuration, voter entry count.

Phase	Scope
Phase 4 - Admin + controls	Reports, audit log, suspicious activity flags, reconciliation, admin adjustments/refunds.
Phase 5 - Enrichment + growth	Social API enrichment, notifications, share optimization, funnel analytics and conversion tuning.

21. Reference flow for developers

	Happy path Nominee page -> Vote -> Login (if needed) -> Choose votes -> Create order -> Pay with SantimPay -> Server confirms payment -> Atomic vote finalization -> 1 raffle entry -> Leaderboard update -> Success page -> Vote Again / Share / Leaderboard.
	Failure path Payment failed/expired -> keep nominee and quantity -> Try Again -> create new payment attempt. Payment status uncertain -> remain pending and reconcile; never award votes until server-confirmed success.
	Refund path Confirmed refund/reversal -> append negative vote ledger entry -> refresh leaderboards -> review/void raffle eligibility according to draw state -> retain full audit history.

END OF SPECIFICATION